Cost-Effective Predictive Modeling for Customer Targeting

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Abstract

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Index Terms: feature selection, profit maximization, customer targeting, ensemble methods

1. Introduction

2. Methodology

2.1. Problem formulation and evaluation

We address cost-aware customer targeting on a labelled training set (X, y), where $X \in \mathbb{R}^{5000\times 500}$ and y is a roughly balanced binary response. Each customer is assigned an acceptance probability $\hat{p} = P(y{=}1 \mid x)$; the campaign contacts the top $k \leq 1000$ customers by decreasing $\hat{p}$. Optimisation targets a profit-oriented score rather than classification accuracy:

$$\text{Score} = 10\,\text{TP} - 5\,\text{FP} - 200\,N_{\text{vars}},$$

where N_{vars} denotes the number of features declared in the submission. The linear penalty on N_{vars} couples predictive performance with model parsimony: each additional variable requires twenty extra true positives to break even.

All design choices—feature subsets, model hyperparameters, and the contact count k—are estimated on training data using 5-fold stratified cross-validation. The business scorer ranks validation-fold probabilities, computes $\text{Score}_k = 10\,\text{TP}_k - 5\,\text{FP}_k - 200\,N_{\text{vars}}$ for $k = 1, \ldots, 1000$, and selects the maximising k. Out-of-fold (OOF) predictions aggregate held-out fold scores and are used for contact-set size selection, thereby avoiding optimistic bias from in-sample fitting. Models are subsequently refit on the full training set prior to test-time ranking. Under the contact cost structure, a customer is profitable in expectation iff $\hat{p} > 1/3$, since $E[\text{contact}] = 15p - 5$.

2.2. Feature selection

Feature selection proceeds in three stages and yields a ranked shortlist together with prefix subsets `top_01`–`top_26` used throughout the modelling pipeline.

In *Stage 1* (500 → 80 features), near-constant columns are removed by a variance threshold, then mutual information and LightGBM split-gain rankings are combined; the 80 columns with the lowest summed rank are retained. *Stage 2* (80 → 26) removes redundancy via correlation pruning ($|r| \geq 0.85$, discarding the lower-MI member of each pair) and iterative variance-inflation-factor filtering (threshold 5), followed by an ℓ_1-regularised logistic path; the penalty parameter C is chosen by cross-validated business score subject to a sparsity band of 20–30 non-zero coefficients.

Stage 3 orders the surviving features by *drop-column cross-validation*. For each feature f, a classifier is evaluated on all Stage-2 columns except f; the quantity $\Delta_f = \text{score}_{\setminus f} - \text{score}_{\text{full}}$ quantifies the marginal profit loss attributable to f (smaller Δ_f indicates higher importance). Prefix subsets of the resulting ordering are rescored under the same metric. Alternative importance measures (mean decrease in impurity, permutation importance, Boruta) are reported for comparison but are not used to define the primary feature order.

2.3. Baselines

Reference models provide lower bounds on performance and verify the implementation of the business scorer; they do not enter the submission pipeline.

Dummy classifiers are fit on a zero-dimensional feature matrix ($n_{\text{features}} = 0$). The `prior` and `uniform` strategies produce constant acceptance probabilities; the `stratified` strategy yields row-wise random one-hot pseudo-probabilities. A `RandomScoreClassifier` assigns independent $\text{Uniform}(0, 1)$ scores across five random seeds. Learned references comprise logistic regression on the highest-correlation univariate predictor (`var_389`) and on the full 500-dimensional input, the latter illustrating the effect of the feature penalty. Supplementary experiments compare prefix subsets of ranked raw features with an equal number of PCA components, evaluated by top-k F1 and ROC-AUC without the variable-cost term.

2.4. Top-k profit curve

The primary modelling strategy selects a single classifier and feature subset by maximising out-of-fold profit. Logistic regression, LightGBM, and XGBoost are compared on each prefix subset under the cross-validated business score; hyperparameters are tuned on the winning configuration using the same objective rather than ROC-AUC. OOF acceptance probabilities are obtained by 5-fold stratified cross-validation, and the contact count is set to $k^* = \arg\max_k \text{Score}_k$ on the resulting profit curve.

2.5. Expected-value profit targeting

This approach retains the model-selection procedure of Section 2.4 but determines k from economically motivated rules when the profit curve is flat near the contact cap. OOF probabilities are mapped through isotonic regression prior to decision-making. Four candidate values of k are derived: (i) a threshold walk that includes customers while $\hat{p} \geq 1/3$; (ii) the smallest k attaining at least 95% of the peak OOF score; (iii) the maximiser of mean fold-wise business score over a discrete grid of k; and

Table 1: *Top-ranked features after drop-column CV (Stage 3).*

Rank	Feature
1	`var_389`
2	`var_308`
3	`var_93`
4	`var_87`
5	`var_254`

(iv) the unconstrained profit-curve maximum. By default, the *combined* rule $k = \min(k_{\mathrm{ev}}, k_{\mathrm{elbow}}, k_{\mathrm{nested}})$ is adopted.

2.6. Rank fusion committee

We examine whether a committee of heterogeneous rankers, combined by OOF-weighted averaging, improves upon any single expert. The committee comprises four fixed members: logistic regression on `top_03`, LightGBM on `top_05`, logistic regression on `top_08`, and a Borda ranker on `top_05` that aggregates feature-wise signed ranks into pseudo-probabilities. Expert weights are proportional to non-negative cross-validated business scores, $w_e \propto \max(0, \mathrm{CV\ business}_e)$, and the fused prediction is $\hat{p} = \sum_e w_e \hat{p}_e$. The submission lists the union of all expert feature sets (eight variables).

2.7. Cluster split targeting

Cluster-split targeting introduces customer heterogeneity through hard partitioning before local estimation. For each prefix subset, the number of KMeans clusters $K \in \{2, \ldots, 10\}$ is assessed via inertia and silhouette; we set K=4 after visual inspection of the diagnostic curves. Within each cross-validation fold, features are standardised, KMeans labels are assigned, and a logistic model is fit per cluster when the training partition contains at least 80 observations; otherwise a fold-level fallback model is used. The same feature set serves both clustering and classification. The optimal prefix subset is selected by comparing OOF business scores across the full sweep.

2.8. Segment-aware targeting

Segment-aware targeting extends the cluster-split framework by separating the feature spaces used for segmentation and for response modelling, and by replacing KMeans with a Gaussian mixture model. In each fold, a GMM with K=5 components is fit to scaled `top_10` features (optionally after PCA); per-segment logistic regressions are estimated on `top_03` only. Variables used for segmentation are treated as latent and are excluded from the submission, which declares three features. Pooled OOF probabilities are ranked and k is chosen by the profit-curve criterion of Section 2.4.

3. Results

3.1. Feature selection

Drop-column ranking places `var_389` first (Table 1). The penalised cross-validated business score is maximised by `top_01` (2 301) and decreases monotonically with subset cardinality (`top_26`: −2 684), whereas ROC-AUC continues to increase (Table 2), indicating a divergence between discrimination and profit under the feature penalty.

Table 2: *Stage-3 prefix subsets: 5-fold CV business score (with feature penalty) and ROC-AUC.*

Subset	N_{vars}	CV business	ROC-AUC
`top_01`	1	**2 301**	0.547
`top_03`	3	1 891	0.551
`top_05`	5	1 490	0.590
`top_08`	8	898	0.583
`top_10`	10	486	0.617
`top_15`	15	−484	0.619
`top_20`	20	−1 472	0.625
`top_26`	26	−2 684	0.627

Table 3: *Reference baseline scores (5-fold CV, with feature penalty).*

Baseline	Features	Business	F1@top-k
Prior / uniform dummy	0	2 475	0.666
Stratified dummy	0	2 474	0.665
Random ranker (mean)	0	2 475	0.665
Logistic (`var_389`)	1	2 301	0.668
Logistic (500 feat.)	500	−97 493	0.668

3.2. Baselines

Table 3 summarises reference scores under 5-fold cross-validation. Dummy classifiers without declared features score approximately 2 475 at k=1000, consistent with the class prior under a balanced sample. Logistic regression on `var_389` attains 2 301 and therefore falls below the no-skill floor once the single-feature penalty is applied. With all 500 variables declared, the best-tuned logistic model yields −97 493 despite F1 $\approx$ 0.668; the corresponding OOF profit curve is shown in Appendix 2.

A supplementary comparison contrasts prefix subsets of Stage-3 ranked raw features with an equal number of PCA components (all 500 columns scaled; 25 components fit once). The same logistic penalty C as the best 500-feature baseline is used. Table 4 reports top-k F1 and ROC-AUC only—no business score or feature penalty, since PCA dimensions are not submission variables. Raw ranked features achieve equal or higher F1 at every k; PCA attains slightly higher ROC-AUC only at k=5 and k=10, but not at k=25. This might indicate that more features introduce noise.

3.3. Top-k profit curve

Among the candidate model families, logistic regression on `var_389` achieves the highest cross-validated business score (2 301; C=0.001 after hyperparameter tuning). The OOF profit curve peaks at k=997 with score 3 290 (565 TP, 432 FP; Table 6, Appendix 3). Tree-based models and enlarged feature subsets fail to improve the penalised objective.

3.4. Rank fusion

The weighted committee attains an OOF score of 2 085 at k=994 when eight features are declared. Although true positives increase to 577 relative to the single-expert logistic model, probability averaging degrades ranking at the head of the list; see Appendix 5.

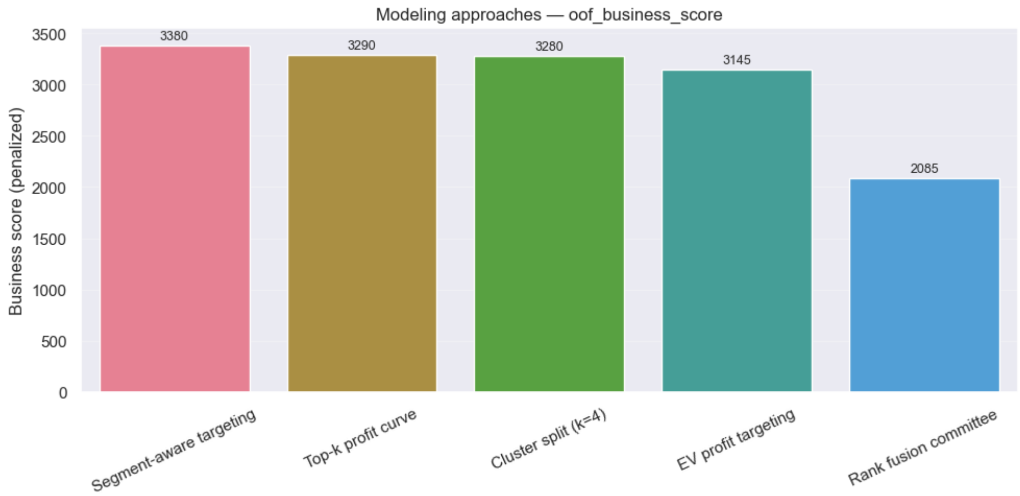

Figure 1: *OOF business score across modelling approaches.*

Table 4: *Top-k raw ranked features vs. PCA components (5-fold CV logistic regression).*

k	F1 raw	F1 PCA	ROC-AUC raw	ROC-AUC PCA
1	**0.668**	0.666	0.555	0.555
3	**0.669**	0.665	0.570	0.584
5	**0.669**	0.665	0.581	0.591
10	**0.669**	0.666	0.594	0.598
15	**0.670**	0.667	0.597	0.599
25	**0.670**	0.667	0.603	0.598

Table 5: *Cluster-split OOF business score by prefix subset (K=4).*

Subset	k	OOF score
`top_05`	998	**3 280**
`top_03`	996	2 880
`top_01`	999	2 800
`top_04`	1 000	2 660
`top_08`	1 000	2 370
`top_10`	999	2 260
`top_26`	996	−895

3.5. Cluster split

Table 5 reports the cluster-split sweep at K=4. The best configuration uses `top_05` (OOF 3 280 at k=998; five features). Enlarging the feature set beyond `top_05` reduces OOF performance; diagnostic plots appear in Appendix 6–7.

3.6. Segment-aware targeting

Segment-aware targeting attains the highest OOF score in the classic pipeline—3 380 at k=998 (598 TP, 400 FP) with three declared features (Table 6, Appendix 4). Relative to cluster split, the confusion counts are marginally less favourable, yet the reduction of two submitted variables (−400 penalty) produces a net gain of 100 points.

Table 6: *OOF results by modelling approach (classic pipeline).*

Approach	Feat.	k	TP	FP	OOF score
Segment targeting	3	998	598	400	**3 380**
Top-k profit curve	1	997	565	432	3 290
Cluster split	5	998	618	380	3 280
EV targeting	1	939	536	403	3 145
Rank fusion	8	994	577	417	2 085

3.7. Approach comparison

Table 6 and Figure 1 aggregate performance across the five modelling strategies. Segment-aware targeting occupies the first rank; rank fusion occupies the last.

4. Conclusions

The experiments indicate that profit-aligned feature selection is a prerequisite for competitive performance: `var_389` dominates Stage-3 subsets, and additional variables reduce the business score despite gains in ROC-AUC. Segment-aware targeting achieves the most favourable balance between OOF profit (3 380) and feature parsimony (three variables). More features means more penalty, which degrades the performance of the committee, but might also introduce more noise (PCA analysis) as F1 did not improve.

5. AI Disclosure

This report was edited with AI assistance, to improve clarity, consistency, and formatting. Part of the docstrings, some phrasing and code refactoring or debugging were refined using AI tools, but all scientific decisions and interpretations are the authors' own. The core content, experimental design, and analysis were developed by the authors.

A. Supplementary figures

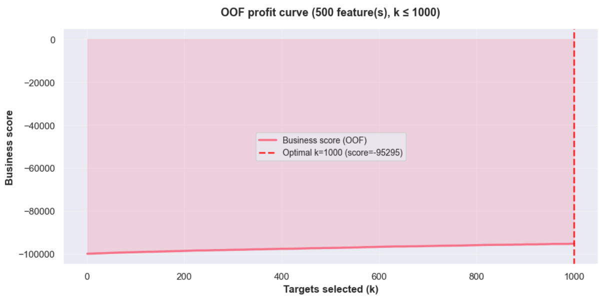

Figure 2: *OOF profit curve for 500-feature logistic regression (TP = 647 at k=1000, score −95 295).*

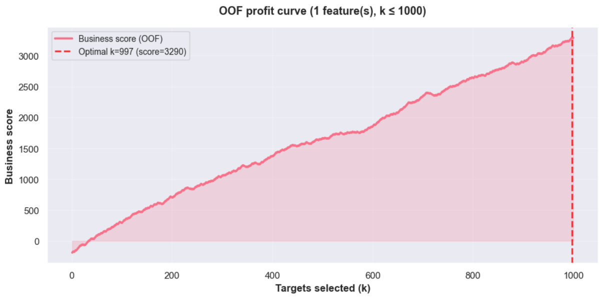

Figure 3: *Top-k profit curve: logistic regression on* `var_389` *(k=997, OOF 3 290).*

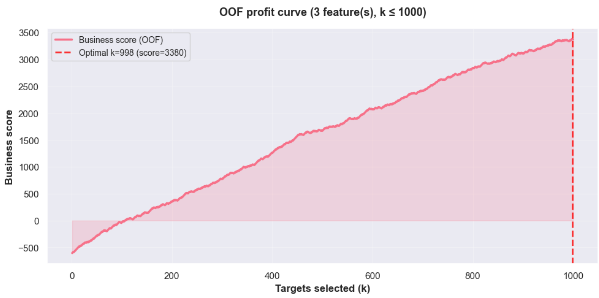

Figure 4: *Segment-aware targeting OOF profit curve (k=998, OOF 3 380).*

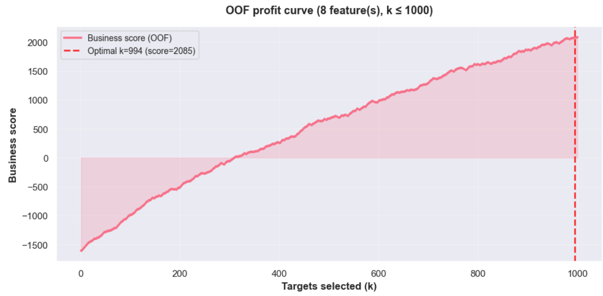

Figure 5: *Rank fusion committee OOF profit curve (OOF 2 085, eight features).*

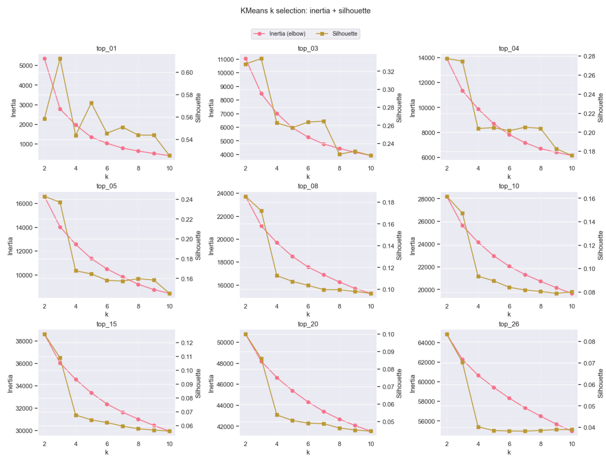

Figure 6: *KMeans diagnostics (inertia and silhouette) for cluster split.*

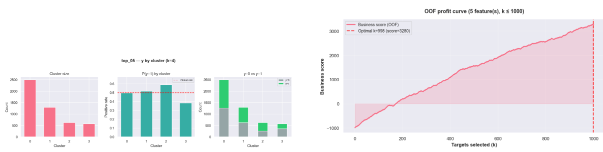

Figure 7: *Cluster split on* `top_05`*: cluster overview (left) and OOF profit curve (right, score 3 280).*

Figure 8: *Cluster split: per-cluster positive-rate profiles.*